Set
Mon 7 Sep 2026

Problem Set 1 — Diagnostic

Session 1 · Course overview and school math refresher

Part A · Algebraic care

1 EXPONENTS

Simplify $\dfrac{27^{2/3} \cdot 3^{-2}}{9^{1/2}}$, stating which exponent law you use at each step.

> **SOLUTION** $27^{2/3} = (3^3)^{2/3} = 3^2 = 9$ by $(a^m)^n = a^{mn}$; $9^{1/2} = 3$. So the expression is $\frac{9 \cdot 3^{-2}}{3} = \frac{1}{3}$.

2 RADICALS

Decide whether $\sqrt{x^2} = x$ holds for all real x. If not, give the correct identity and a value of x that breaks the false one.

> **SOLUTION** False. $\sqrt{\cdot}$ denotes the non-negative root, so $\sqrt{x^2} = |x|$. At $x = -3$: $\sqrt{9} = 3 \neq -3$.

3 IDENTITIES

Prove that $(a+b)^2 - (a-b)^2 = 4ab$ for all real a, b. State explicitly why your argument covers *all* real a, b and not just the ones you tried.

> **SOLUTION** Expand both squares: $(a^2 + 2ab + b^2) - (a^2 - 2ab + b^2) = 4ab$. Every step is a ring identity valid for arbitrary reals, so no case analysis is needed — the letters were never assumed to be particular numbers.

4 COMPLETING THE SQUARE

Write $2x^2 - 12x + 23$ in the form $a(x-h)^2 + k$ and read off its minimum value and where it occurs.

> **SOLUTION** $2(x^2 - 6x) + 23 = 2\big((x-3)^2 - 9\big) + 23 = 2(x-3)^2 + 5$. Since $(x-3)^2 \geq 0$ with equality only at $x = 3$, the minimum is 5 at $x = 3$.

Part B · Absolute value and inequalities

5 ABSOLUTE VALUE

Solve $|3x+1| \geq 4$. Mark each step $\Longleftrightarrow$ or $\Longrightarrow$ and say why.

> **SOLUTION** $|u| \geq 4 \Longleftrightarrow u \leq -4$ or $u \geq 4$ (definition of $|\cdot|$, a case split, not a loss of information). With $u = 3x+1$: $3x \leq -5$ or $3x \geq 3$, i.e. $x \leq -\frac{5}{3}$ or $x \geq 1$. Dividing by $3 > 0$ preserves the direction, so every step is an equivalence.

6 RATIONAL INEQUALITY

Find all real x with $\dfrac{2}{x-1} \leq 1$. Handle the sign of $x-1$ explicitly; do not multiply through by an expression of unknown sign.

> **SOLUTION** Case $x > 1$: multiplying by $x-1 > 0$ gives $2 \leq x-1$, so $x \geq 3$. Case $x < 1$: multiplying by $x-1 < 0$ flips the inequality, $2 \geq x-1$, i.e. $x \leq 3$, which holds throughout $x < 1$. At $x = 1$ the expression is undefined. Answer: $x < 1$ or $x \geq 3$.

7 AM–GM

Show $x^2 + y^2 \geq 2xy$ for all real x, y, and determine exactly when equality holds.

> **SOLUTION** $x^2 + y^2 - 2xy = (x-y)^2 \geq 0$ because squares of reals are non-negative. Equality holds iff $(x-y)^2 = 0$, i.e. $x = y$.